RESEARCH MEETING / PAPER SUMMARY

HBF Sucks?

A Full-Stack Characterization of High-Bandwidth Flash for KV-Centric LLM Serving

A faster flash tier can make the serving system slower.

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Why Replace SSD with HBF?

MOTIVATION / THE INTUITIVE UPGRADE

KV cache grows beyond GPU memory

Serving systems offload reusable KV to a larger backing tier.

SSD supplies capacity

But restoring KV incurs storage access latency.

HBF promises a faster flash tier

Package-local flash offers lower latency and higher bandwidth than SSD.

Does faster flash translate into faster end-to-end LLM serving?

The paper tests the intuitive "SSD → HBF" upgrade at the full-system level.

Source: HBF Sucks?, abstract and introduction.

What Did the Paper Actually Evaluate?

METHOD / DROP-IN SUBSTITUTION

Keep the Mooncake-style runtime
Admission / retention / lookup / KV save & restore

Swap capacity-matched SSD for HBF-1 / HBF-2

Extended TokenSim
4 complete two-hour traces; 5 dense / MoE models
H100 / B200 profiles; thermal + endurance models

HOW THE MODELED ORGANIZATIONS DIFFER

	HBF-1	HBF-2
Package stack sites	All assigned to HBF	Split equally: HBM + HBF
GPU near tier	GDDR7 on the card PCB	HBM3e on the interposer
HBF capacity / bandwidth	2× HBF-2	Reference (1×)

At the same GPU profile, near-tier capacity is equal; memory technology and bandwidth differ.

Same SSD-style runtime, but HBF trades near-tier resources for package-local flash.

Source: Sec. 2.2, Tables 2 and 3(c), Sec. 6.1. Modeled roadmap profiles, not production hardware.

Three Conditions for HBF to Help

ORGANIZING PRINCIPLE

C1 — Critical Path

Read I/O must be a real serving bottleneck.

F1–F2: package tradeoff and limited base-die compute leverage.

C2 — Reuse

Useful reads must repay every HBF write.

F3–F4: cold write-heavy traffic and limited batching benefit.

C3 — Sustainability

High bandwidth must remain sustainable.

F5–F6: thermal throttling and flash write wear.

6 findings = 2 findings for each condition

A faster medium helps only if all three conditions hold.

Source: Sections 5–8.

Faster HBF Barely Improves E2E Latency

C1 FAILS / FLASH IS BARELY ON THE CRITICAL PATH

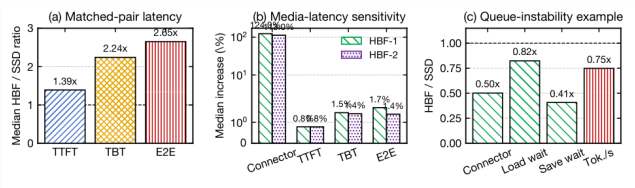


Fig. 4: matched pairs, media sweep, and queue behavior.

Less near-tier capacity/bandwidth
→ fewer admitted requests
→ more queueing
→ worse E2E latency

EVIDENCE

Faster connector, yet mean E2E latency is
2–5.5× higher
than capacity-matched SSD.

3.75× faster raw HBF latency
→ only **0.75% / 0.88%** lower E2E
HBF-1 / HBF-2, vs. their slower-media settings.

→ exposed secondary-tier I/O ≈ **1%**

The bottleneck is not flash-media latency.

Source: Fig. 4 and Sec. 6.1. HBF-1 vs. SSD24; HBF-2 vs. SSD12. H100/B200 simulation.

All Production Traces Send More Writes Than Reads to HBF

C2 FAILS / USEFUL READS DO NOT REPAY WRITES

Metric	traceA	traceB	coder	thinking
Write/read ratio	2.14×	4.90×	1.14×	2.20×
ρ (reads/write)	0.47	0.20	0.88	0.45
Prefix hit rate	38.21%	52.46%	45.58%	44.12%

Table 8: flash-tier byte traffic over each full two-hour replay.

Hot, highly reused KV → retained in HBM / GDDR

Cold, low-reuse KV → evicted to HBF

EVIDENCE / WRITE BYTES PER READ BYTE

traceA **2.14**×

traceB **4.90**×

coder **1.14**×

thinking **2.20**×

All exceed **1**× **(equal write/read bytes)**. Every write costs flash work; many blocks are never read back.

The cache hierarchy gives HBF a write-heavy, low-reuse KV stream.

Source: Table 8 and Sec. 7.1. Production-trace replay, not HBF device measurements.

Write-Heavy KV Traffic Makes Peak HBF Bandwidth Unsustainable

C3 FAILS / THERMAL LIMITS AND WRITE WEAR

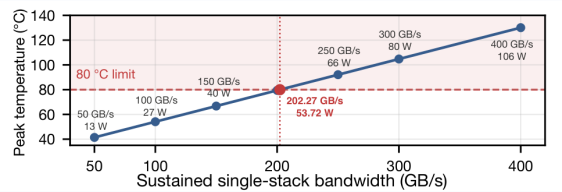


Fig. 10(a): modeled temperature vs. single-stack bandwidth.

Thermal = short-term throttling

16-Hi TLC HBF model; **80°C** limit

$\approx$ **202.27 GB/s** sustainable point

Peak interface BW $\neq$ sustained workload BW

EVIDENCE / LONG-TERM ENDURANCE

Endurance = long-term write wear

Measured trace writes:

48–140 TB/day

Modeled HBF lifetime:

$\approx$ 0.56$\times$

the capacity-matched SSD pool

Under unchanged KV pooling and the paper's TLC write-budget model.

Write-heavy KV reduces sustained bandwidth and consumes flash lifetime.

Source: Fig. 10(a), Fig. 11 and Sec. 8. Model projections, not hardware lifetime measurements.

The Device Is Fine; the Drop-In Deployment Is Not

CONCLUSION / PLACEMENT REQUIREMENTS

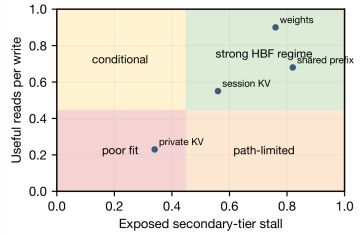


Fig. 12: conceptual map; not a calibrated boundary.

Poor HBF targets

Private KV
Transient decode KV
Low-reuse objects

Better HBF targets

Model weights
Shared prefixes
Read-mostly objects

Requirements for a future runtime

Critical-path-aware / reuse-aware
Write-budgeted / thermal-aware

HBF should not be treated as a faster SSD.

The paper characterizes the problem and gives design requirements; it does not implement a complete new HBF-aware serving runtime.

Source: Fig. 12 and Sections 9–10.

Sources Behind the HBF Evaluation

SYSTEM / WORKLOAD

TokenSim / HBF code

pku-lemonade research project
Open-source serving simulator
→ Serving, KV transfers, queueing and E2E latency

Qwen-Bailian traces

Alibaba / SJTU; production data
→ Request and reuse replay: traceA, traceB, coder, thinking

HBF DEFINITION

OCP HBF v0.7.0

Open Compute Project — **industry spec**
Contributors: Sandisk + SK hynix
→ Interface, page-write and lifecycle rules

KAIST HBF roadmap

Joungho Kim / KAIST TERA Lab
→ HBF-1 / HBF-2 organizations
Academic roadmap, not a product datasheet

HAVEN

Hsu et al.; academic architecture/model
→ NAND latency, energy, bandwidth, power

PHYSICAL LIMITS

3D-ICE

EPFL; open-source thermal tool
→ 16-Hi HBF temperature model
80°C policy: chosen by this paper.

KIOXIA CM7-V data

KIOXIA America; vendor datasheet
→ SSD baseline / rated endurance
HBF lifetime: authors' TLC budget + trace writes; not a wear-out test.

2028 / 2030 are roadmap generations, not guaranteed mass-production dates.

No physical HBF hardware is measured here: the full-stack model combines code, specifications, roadmaps, device models, production traces and thermal modeling.

Source: Paper Refs. [1], [9], [11], [13], [23], [24], [26]; Sections 2 and 4.